ABOUT:

This record contains data on commodity-driven deforestation and associated carbon emissions for the period 2001-2023, estimated by the Deforestation Driver and Carbon Emission (DeDuCE) model (Singh & Persson, 2026). DeDuCE combines remote sensing data on forest loss and land-use with agricultural statistics to identify and attribute deforestation across the world to the expansion of cropland, pastures and forest plantations, and the commodities produced on this land. This file also contains data on deforestation embodied in the production, exports, imports, and consumption of agricultural and forestry commodities by country, year, and commodity for the time period 2005-2023, derived using a physical and a hybrid (physical-monetary) trade model.

More information about the DeDuCE model, including input data, model code, and data outputs, can be found here: https://www.deforestationfootprint.earth/ (online dashboard, hosting all the data, allowing users to interact with the data, by querying, downloading and visualising both the deforestation and trade data)

Openly accessible DeDuCE model code: https://github.com/chandrakant6492/DeDuCE

FILES:

Excluding 'About the data.pdf' and 'ReadMe.txt', there are three files in this repository:

1. **DeDuCE_Deforestation attributed to agricultural and forestry commodities (2001-2023).xlsx:**

   Presents the attribution of embodied deforestation area (in ha/yr) and associated net carbon emissions from changes in above-ground, below-ground and soil carbon, incl. peat drainage (in tCO2/yr), by country of production, year and commodity (note that this excludes any forest loss that was not attributed to expanding pastures, cropland or forest plantations) (Singh & Persson, 2026). Data are provided both for the attribution of forest loss to different land uses and associated (committed) emissions in each year (unamortized), and for the amount of deforestation and emissions embodied in the production of associated commodities (amortised over 5 years, available from 2005 onwards). Instances of net negative deforestation emissions reflect situations where the carbon stock of the replacing land use (e.g. tree crops) exceeds the gross carbon loss from forest clearing.

2. **Physical trade model_Deforestation embedded in global supply chains (2001-2023).csv:**

   Presents estimates of bilateral links between countries of production and countries of apparent consumption for commodities driving deforestation. Values are presented in terms of amortised deforestation area (in ha/yr) and associated net emissions from losses in above-ground, below-ground and soil (incl. peat drainage) carbon (in MtCO2/yr). The data

is produced using the trade model developed by Kastner et al. (2011) and updated by Schwarzmueller and Kastner (2022), which establishes links between countries of primary production and countries of apparent consumption (in primary or processed form). The calculations rely on production and bilateral data on physical trade flows available in FAOSTAT (as available in January 2024). As FAOSTAT does not provide bilateral trade data for leather and wood products, these items are excluded from this dataset. This approach considers the physical flows of agricultural products in their primary form and up to the level of processing covered in FAOSTAT's trade data (e.g. flour, fruit juices). Traded products are converted into their primary equivalents of the agricultural commodity they are matched to (for details and factors, refer to Schwarzmueller and Kastner, 2022). The values here do not include estimates of products indirectly embodied in the production of other products (e.g., feed in meat products, biofuels used in industrial processes). To limit the size of the dataset, we include the links cumulatively making up (per year) 99.9% of the total positive values of attributed deforestation area (the full dataset can be obtained from the authors).

3. (New) **Hybrid trade model_Deforestation embedded in global supply chains (2001-2023).csv:**

   Presents the embodied deforestation area (in ha/yr) embedded in total consumption, and associated net carbon emissions from losses in above-ground, below-ground and soil, incl. peat drainage (in $MtCO_2\ yr^{-1}$), by country or territory of production, country or region of consumption, year and commodity. The data are produced using the hybridised multi-regional input-output (MRIO) framework IOTA (Croft et al. 2018), which in this application utilises national-level production and trade data from FAO and UN COMTRADE, and a global economic MRIO (GLORIA, Lenzen et al. 2017), to link production and associated impacts to localised final consumption. The specific methods utilised in the IOTA application presented here are the same as those used for the Global Environmental Impacts of Consumption indicator (Croft et al. 2023; www.commodityfootprints.earth). To limit the size of the dataset, we include the links cumulatively making up (per year) 99.9% of the total positive values of attributed deforestation area (the full dataset can be obtained from the authors).

CITATION, CONTACT & COPYRIGHT:

Chandrakant Singh, U. Martin Persson, Simon Croft, Thomas Kastner & Chris West (2026). Commodity-driven deforestation, associated carbon emissions and trade 2001-2023. Chalmers University of Technology, Senckenberg Biodiversity and Climate Research Centre, and Stockholm Environment Institute York (SEI-York), York University.
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For inquiries regarding this dataset, contact Martin Persson and Chandrakant Singh by sending an email to deduce.env@chalmers.se.

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UPDATE DETAILS:

The data presented here is an update of the original DeDuCE data (Singh et al. 2024), extending the data series to 2023 and providing higher country resolution for the hybrid trade model data.

CAUTIONS AND NOTES:

While the analysis of commodity-driven deforestation presented here is, to a large extent, derived from spatially explicit data, most commodity-specific deforestation (carbon) footprints are based on a land-balance model implemented at the national level (with the exception of Brazil, where it is run at the municipal level). This implies that the reported deforestation footprints are a mix of attribution based on direct land-use change (dLUC; i.e., commodities produced on the deforested land) and statistical land-use change (sLUC; i.e., commodities expanding in a geographical area, which may happen directly on forest land or indirectly causing deforestation by expanding on other land uses, pushing these into forests). The reported quality score can be used as a proxy to differentiate between dLUC and sLUC estimates, with the former having a quality score above 0.6. Note also that the deforestation attribution to trade flows and consumption reflects national averages, not reflecting differences in geographical sourcing between importing countries. The estimates of deforestation embodied in imports should therefore be interpreted as a measure of deforestation risk exposure for a given importing country.

Note that the two trade models used to assess deforestation and the associated carbon emissions embodied in consumption differ in their system boundaries, structures, and input data, implying that the results are not directly comparable. The physical trade model covers direct trade in agricultural commodities across countries but does not include highly processed products (especially non-food products), or the indirect flows of embodied deforestation in other products that use these agricultural products as inputs (e.g., meat produced using soy feed or biodiesel produced from palm oil). Consequently, the data from the physical trade model provide an estimate of the direct impact of consumption on trade-driven deforestation. The hybrid multi-regional input-output model (IOTA), on the other hand, includes not only direct trade of commodities linked to deforestation, but also trade and consumption of all products and services that use these commodities as inputs, along complex international supply chains, using data on monetary transactions between economic sectors. As such, the IOTA model gives an estimate of the total consumption-based deforestation (carbon) footprint of different countries. For a general discussion on the differences between physical trade models and monetary input-output models, see Hubacek & Feng (2016); for a discussion on the differences specifically for deforestation, see Pendrill et al. (2019).

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Singh, C., & Persson, U. M. (2026). Global patterns of commodity-driven deforestation and associated carbon emissions. Nature Food, 7(2), 138-151. https://doi.org/10.1038/s43016-026-01305-4

VERSION HISTORY:

- V1.0 (2020-11-05): Initial dataset (https://zenodo.org/record/4250532)
- V1.1 (2022-01-13): https://zenodo.org/records/5886600
- V2.0 (2024-02-26): Major update, presenting the first dataset based on the DeDuCE model (https://zenodo.org/records/10633818)
- V2.1 (2026-03-11): Major update, switched the MRIO framework from EXIOBASE to GLORIA; replaced the Curtis et al. 2018 forest loss driver dataset with the newer Sims et al. 2025 dominant driver dataset.